

Neighborhood Park Preliminary Design Masterplan

The plan at scale. Every room struck off the crescent's centre; every area the measured area of the drawn polygon.

02

OF 12 UPLOAD SLOTS

15,000 m²

SITE AREA

44.5% → 64.6%

COMFORTABLE DAYLIGHT HOURS

−7.13 °C

HEAT INDEX UNDER CANOPY

131

TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL

wasim437.github.io/AI_SAFA/

01

WHAT IS IN THIS FILE

Phase5 Masterplan Development Report

PLAN

Fig10 masterplan

PLAN

Planting

DRAWING

02

THE SCALE OF THE ANALYSIS BEHIND IT

39 years

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours

of sun position ray-traced against 131 trees for every hour of the year

15,000

one-metre ground cells modelled for shade and comfort, site-wide

4 models

trained and tested for leakage — two of them decide what this document reports

41 checks

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository github.com/wasim437/AI_SAFA

Live portal wasim437.github.io/AI_SAFA/

Produced by **CODE** `src/plan.py` `src/figures.py` **DATA** `data/raw/site_zoning_schedule.csv`

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

02

OF 12

01 THE TEN PHASES

- P1 Site & Context Analysis**
39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk
- P2 Problem Definition**
Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else
- P3 Opportunity & Objectives**
The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against
- P4 Concept Development**
Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage
- P5 Masterplan Development** ★ THIS DOCUMENT
Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²
- P6 Detailed Design**
The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy
- P7 Performance & Sustainability**
Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone
- P8 User Experience & Activation**
K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn
- P9 AI Workflow & Visualisation**
The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film
- P10 Submission Assembly**
Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Plan
src

Figures
src

Site zoning schedule
data/raw

How this document was produced

03

REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py          # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline   # 41 checks
```

Preliminary Design Masterplan

The room schedule, and how 15,000 m² is allocated

The masterplan is generated, not drawn. Every room is a box in the crescent's own polar frame, mapped to site metres and clipped to the boundary — so each area is the measured area of the polygon actually drawn, and the schedule closes on 15,000 m² without anyone reconciling a spreadsheet.

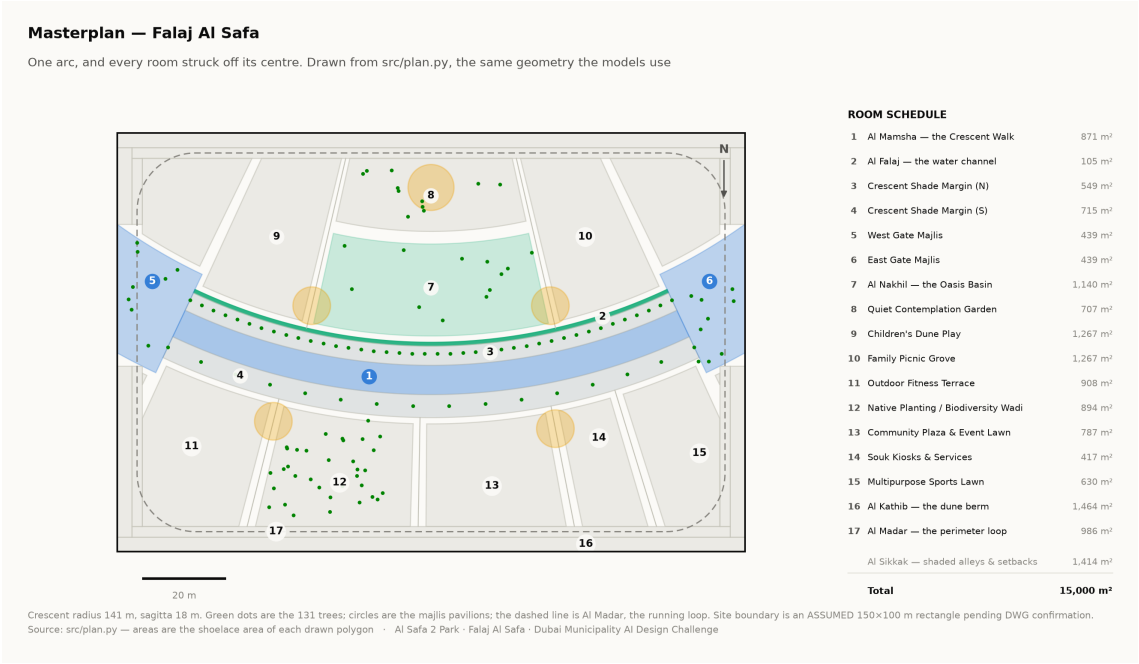
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The organising geometry

One arc. A crescent of shade 141 m in radius sweeps across the site, bowing convex south so that its hollow opens to the north. A 0.9 m water channel runs beneath its northern drip line. Every room in the park is struck off the same centre, so no room is a rectangle and every room faces the crescent square-on.

The arc has a chord of 138 m and a sagitta of 18 m, giving a radius of 141.25 m and a true arc length of 144.2 m. Those four numbers generate everything else in this report.



Masterplan with the numbered room schedule. Technical drawing — to scale.

2. Room schedule

Room	Category	Area (m²)	% of site
Al Mamsha — the Crescent Walk	Circulation	871	5.81%
Al Falaj — the water channel	Water	105	0.70%
Crescent Shade Margin (N)	Green	549	3.66%
Crescent Shade Margin (S)	Green	715	4.77%
West Gate Majlis	Arrival	439	2.93%
East Gate Majlis	Arrival	439	2.93%
Al Nakhil — the Oasis Basin	Green	1,140	7.60%
Quiet Contemplation Garden	Passive	708	4.72%
Children's Dune Play	Active	1,267	8.45%
Family Picnic Grove	Passive	1,267	8.45%
Outdoor Fitness Terrace	Active	908	6.06%
Native Planting / Biodiversity Wadi	Green	894	5.96%
Community Plaza & Event Lawn	Social	788	5.25%
Souk Kiosks & Services	Commercial	417	2.78%
Multipurpose Sports Lawn	Active	630	4.20%

Room	Category	Area (m ²)	% of site
Al Kathib — the dune berm	Green_Buffer	1,464	9.76%
Al Madar — the perimeter loop	Circulation	986	6.57%
Al Sikkak — shaded alleys & setbacks	Circulation	1,414	9.43%

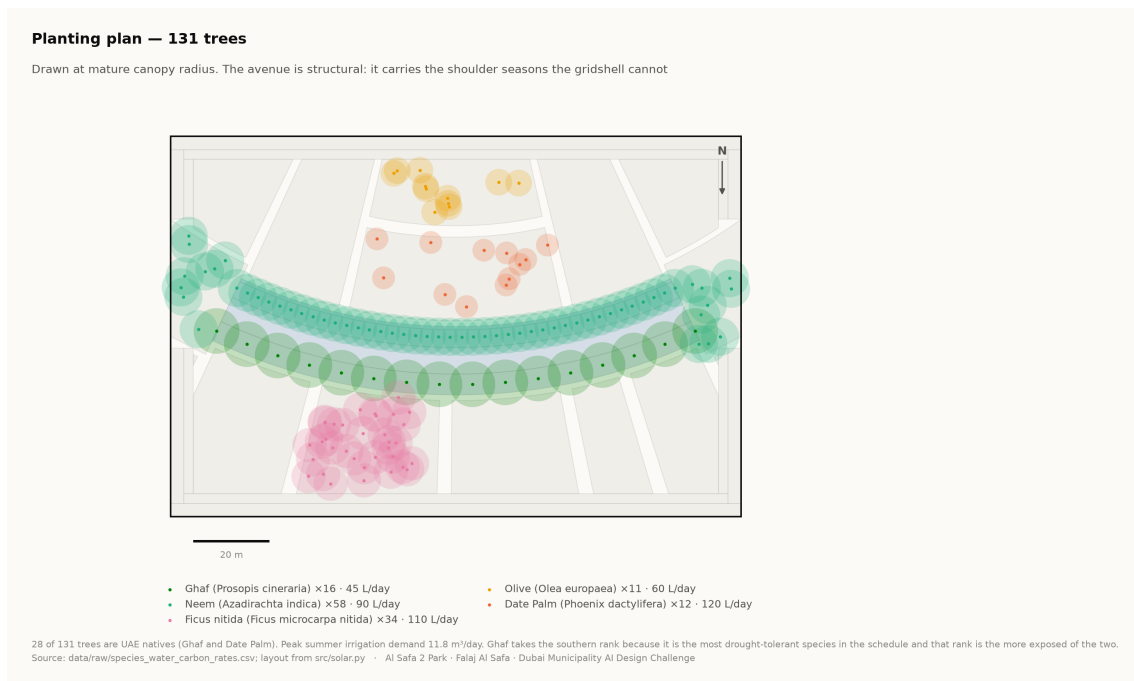
Total 15,000 m². Areas are shoelace areas of the drawn polygons, and the pipeline asserts that the schedule sums to the site area.

3. Where things are, and why

- **The hollow holds the lingering.** The Oasis Basin, the quiet garden, the children's play and the picnic grove all sit on the concave side, which the comfort model shows is measurably cooler than the convex face.
- **The convex face takes the active and the commercial.** The sports lawn, the community plaza and the souk sit south of the arc, where exposure matters less and evening use dominates.
- **The alleys are radii.** Every sikka is a radius of the same circle, so every room presents its face to the crescent rather than a corner.
- **The perimeter is a berm, not a fence.** Al Kathib is planted earth against the roads, doing three jobs at once — noise, glare and heat.

4. Site boundary — a stated assumption

The site is modelled as a 150 × 100 m rectangle. This is an **assumption** pending confirmation against the CAD file issued with the competition documents. Every area figure in this submission depends on it, and it is flagged here rather than left for a juror to discover.



Planting plan — 131 trees at mature canopy radius. Technical drawing — to scale.

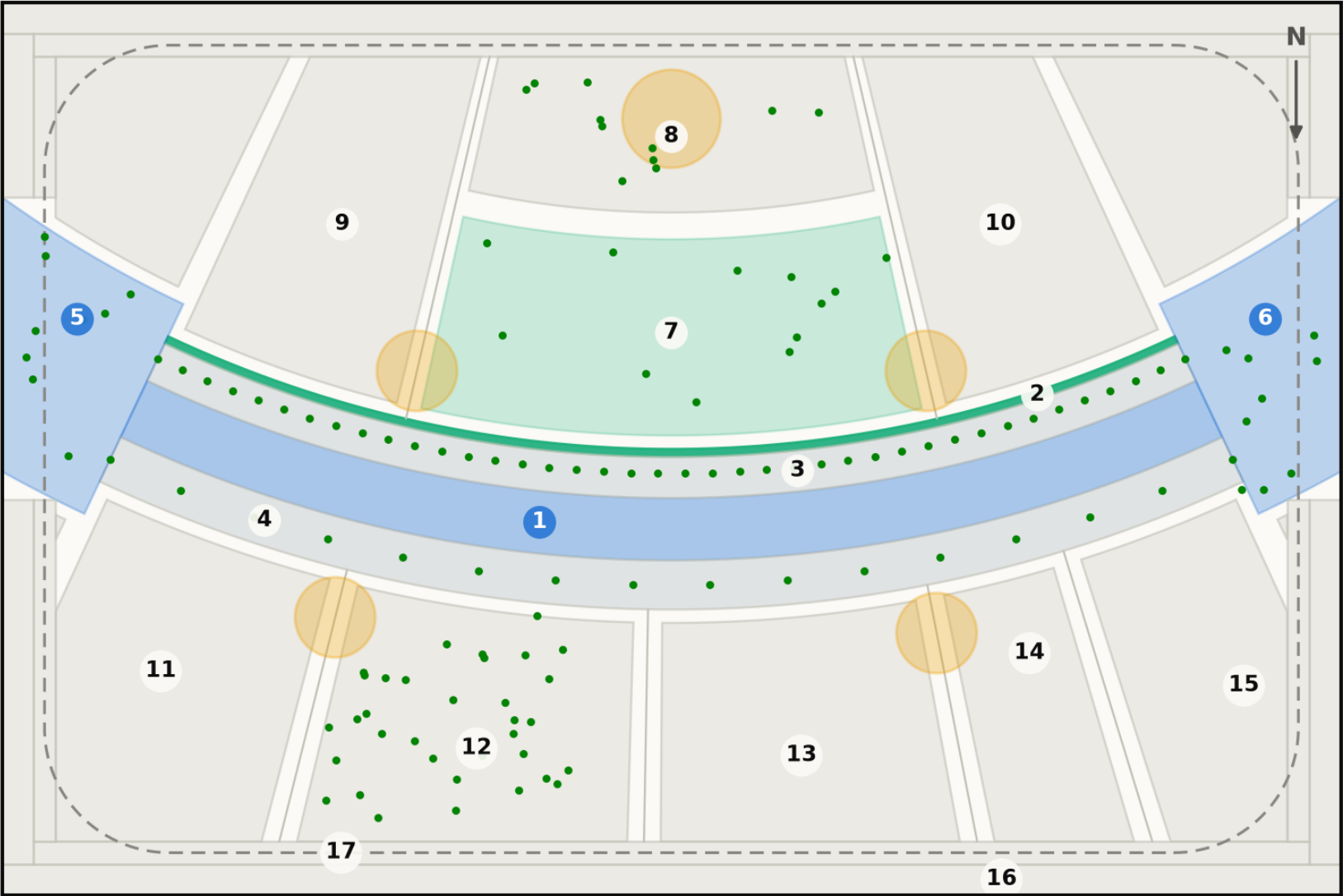
Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²

Total 15,000 m²

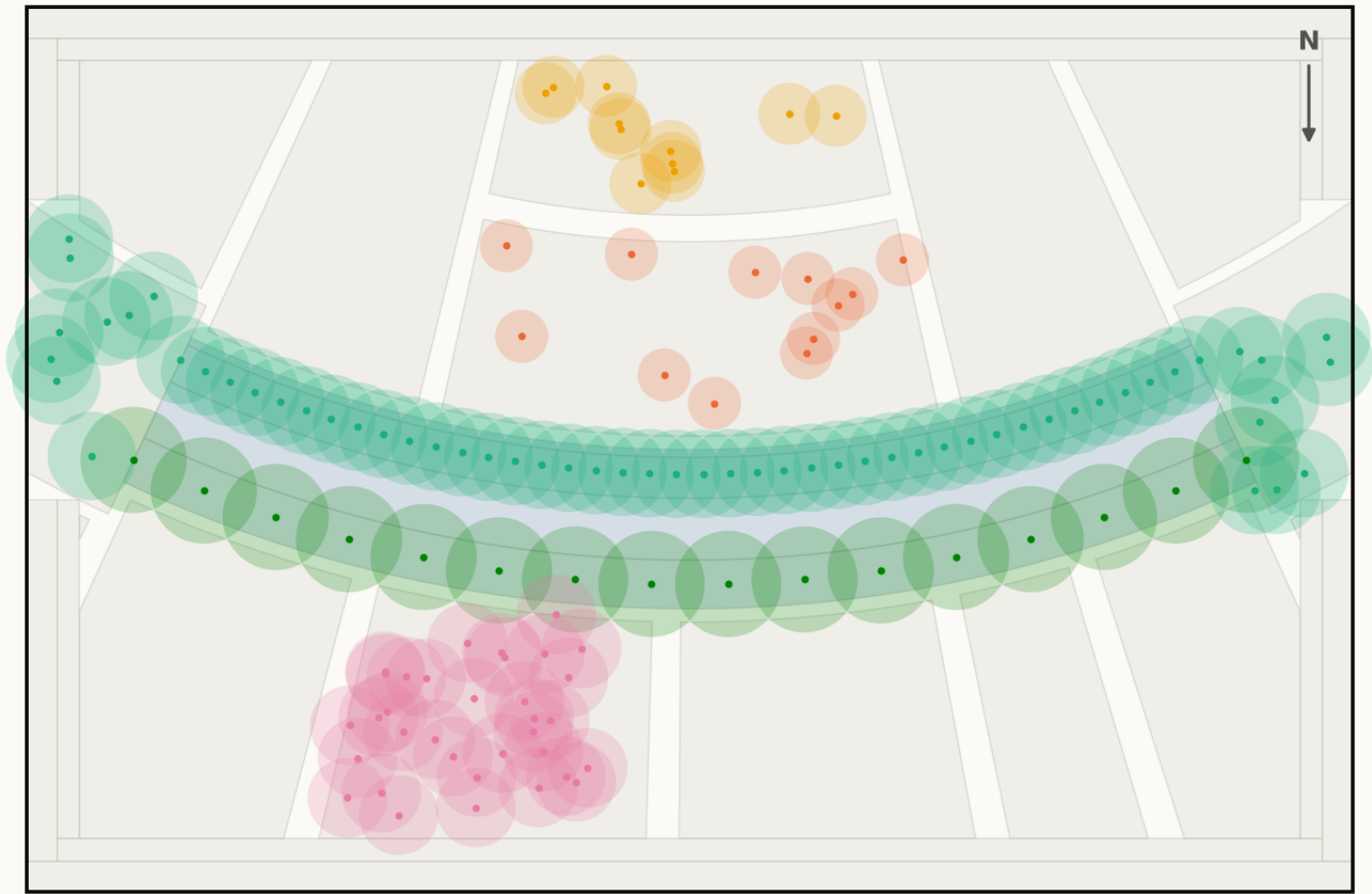
Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting

Technical drawing — to scale.

Planting plan — 131 trees

Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot



- Ghaf (*Prosopis cineraria*) ×16 · 45 L/day
- Olive (*Olea europaea*) ×11 · 60 L/day
- Neem (*Azadirachta indica*) ×58 · 90 L/day
- Date Palm (*Phoenix dactylifera*) ×12 · 120 L/day
- Ficus nitida (*Ficus microcarpa nitida*) ×34 · 110 L/day

28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.
Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Every drawing and every chart in this project

All 18 images the pipeline produces, whichever slot they belong to. Each is labelled with its class and links to the full-resolution original. Nothing here is hand-drawn.

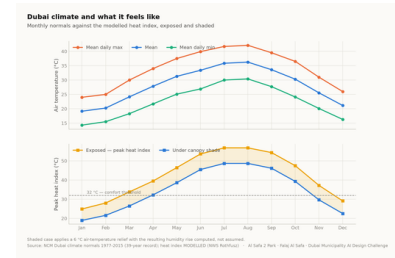


Fig01 climate and comfort
analysis output

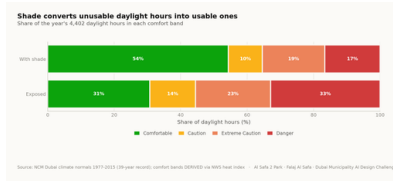


Fig02 comfort bands
analysis output

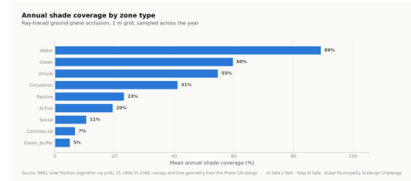


Fig03 shade by zone
analysis output

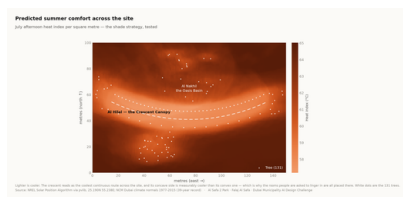


Fig04 site comfort map
analysis output

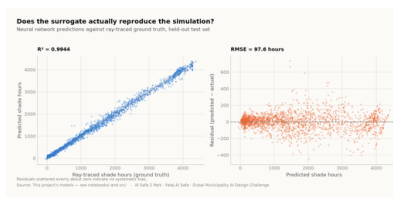


Fig05 surrogate performance
analysis output

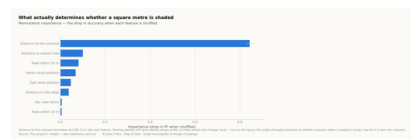


Fig06 feature importance
analysis output

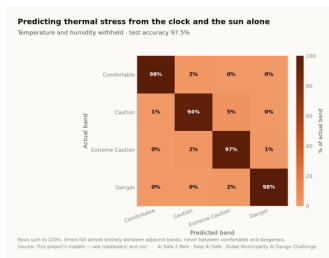


Fig07 confusion matrix
analysis output

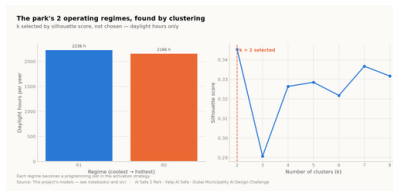


Fig08 microclimate regimes
analysis output

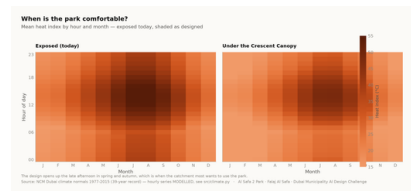


Fig09 diurnal comfort
analysis output



Fig10 masterplan
analysis output

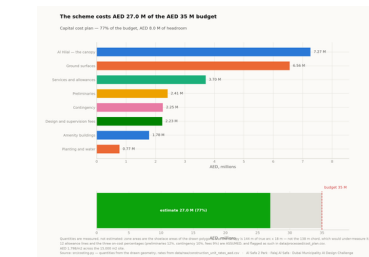


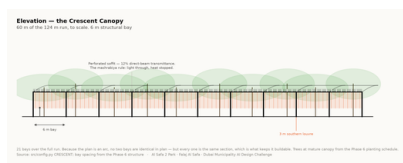
Fig11 cost plan
analysis output



Circulation

technical drawing

The visual record — continued

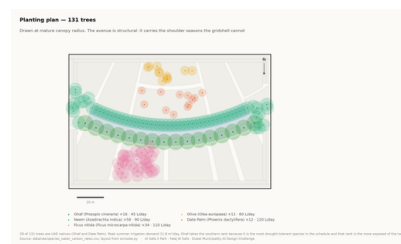


Elevation
technical drawing



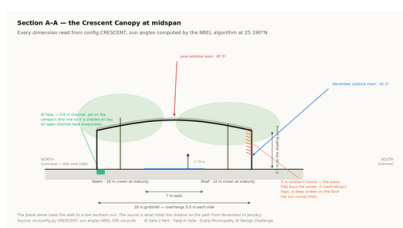
Facilities

technical drawing

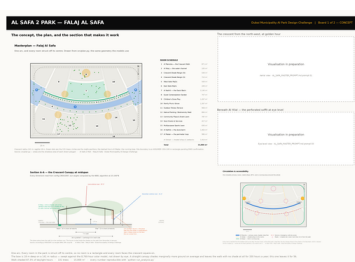


Planting

technical drawing



Section
technical drawing



Board 1 concept
presentation board



Board 2 evidence
presentation board

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

MEASURED RESULT	models/headline_metrics.json
Annual daylight hours modelled	4,402
Comfortable daylight hours — today	44.5%
Comfortable daylight hours — as designed	64.6%
Mean heat-index reduction under canopy	7.13 °C
Peak heat index, exposed	56.8 °C
Peak heat index, shaded	48.7 °C
Crescent Walk shaded, canopy and louvre	87.3%
Site-wide mean shade	34.1%
Trees planted	131

TRAINED MODELS, ON HELD-OUT TEST SETS	models/model_metrics.json
M1a Random Forest — shade surrogate, test R ²	0.9982
M1b Neural network — deployed surrogate, test R ²	0.9944
M2 Gradient Boosting — comfort band, test accuracy	97.5%
M3 K-Means — regimes selected by silhouette	k = 2
Training / validation / test split	10,500 / 2,250 / 2,250
Random seed, fixed for reproducibility	42

RUN IT YOURSELF

```
git clone https://github.com/wasim437/AL_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js   # 64 portal checks
node tests/test_film.js          # every frame of the film
```

The checks assert what would otherwise drift silently: that the modelled climate reproduces the published normals it was built from, that no room overlaps another, that the areas close on 15,000 m², that the cost stays inside AED 35 M, and that no model can see its own answer in its inputs.

The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (7)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

tools/__init__.py	Build tool
tools/build_docs.py	Build tool
tools/build_links.py	Build tool
tools/build_notebook.py	Build tool
tools/build_reports.py	Build tool
tools/build_submission_pdfs.py	Build tool
tools/organize_repo.py	Build tool
tools/report_content.py	Build tool
tools/sync_film.py	Build tool
tools/sync_portal.py	Build tool
tools/sync_submission.py	Build tool

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset
data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (1)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
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